

Project G8: CNNs vs Vision Transformers — Pretraining and Data Augmentation on PASCAL VOC

- **Milestone II status:**

- You completed preliminary baselines for ResNet-50 and ViT-B/16 under both pretrained and from-scratch conditions on the full VOC training set with standard augmentation (Table 1).
- Key results: Pretrained ViT-B/16 achieves the highest mAP (0.9274), pretrained ResNet-50 close behind (0.8938). From-scratch ViT-B/16 triggers early stopping at epoch 8 with mAP=0.1659, while ResNet-50 reaches 0.3092 by epoch 30—this powerfully demonstrates the data-efficiency gap.
- The from-scratch ViT failure is exactly the finding the project was designed to test. Frame this prominently.
- ConvNeXt-T and DeiT-S baselines are not yet run. These are needed for the 4-architecture comparison.

- **Overall:** Good preliminary results that validate the core hypothesis. For the final report: (i) extend baselines to ConvNeXt-T and DeiT-S, (ii) run the data-scaling experiment (5%–100%), (iii) run the augmentation ablation (none/standard/strong).

Core scientific questions for the final report

- RQ1: At what data fraction does ViT (pretrained) match or exceed CNN (pretrained) performance on multi-label classification?
- RQ2: How much does strong augmentation (RandAugment) close the from-scratch data-efficiency gap for ViTs?
- RQ3: Does the CNN-to-transformer modernization in ConvNeXt close the gap with ViT without requiring transformer-scale pretraining?
- Define the centerpiece figure: mAP (y-axis) vs data fraction (x-axis), with separate lines for each of the 4 architectures $\times$ 2 pretraining conditions.

Remaining experiments

- **ConvNeXt-T and DeiT-S baselines:** Run both on full data, pretrained and from-scratch. These complete the 4-architecture comparison.
- **Data-scaling experiment:** For all 4 architectures $\times$ 2 pretraining conditions, train on 5%, 10%, 20%, 50%, 100% of training data (using iterative stratification to preserve multi-label class frequencies). This is 40 configurations; with early stopping at low fractions, most runs will be fast.
- **Augmentation ablation:** For pretrained ResNet-50 and pretrained ViT-B/16, compare: (a) no augmentation, (b) standard (flip + color jitter), (c) strong (RandAugment). Hypothesis: augmentation helps ViT more than CNN.
- **Per-class AP analysis:** Report AP for all 20 classes. Investigate whether ViTs outperform CNNs on “context-dependent” classes (dining table, sofa) while CNNs excel on “texture-defined” classes (cat, dog).

Writing the final report

- **Main message:** e.g., “ViTs achieve the highest multi-label mAP when pretrained, but require $\geq X\%$ of the training data to match CNN performance from scratch. Strong augmentation partially closes this gap. ConvNeXt-T matches ViT performance while maintaining CNN-level data efficiency, suggesting that transformer-era training recipes matter more than architecture.”